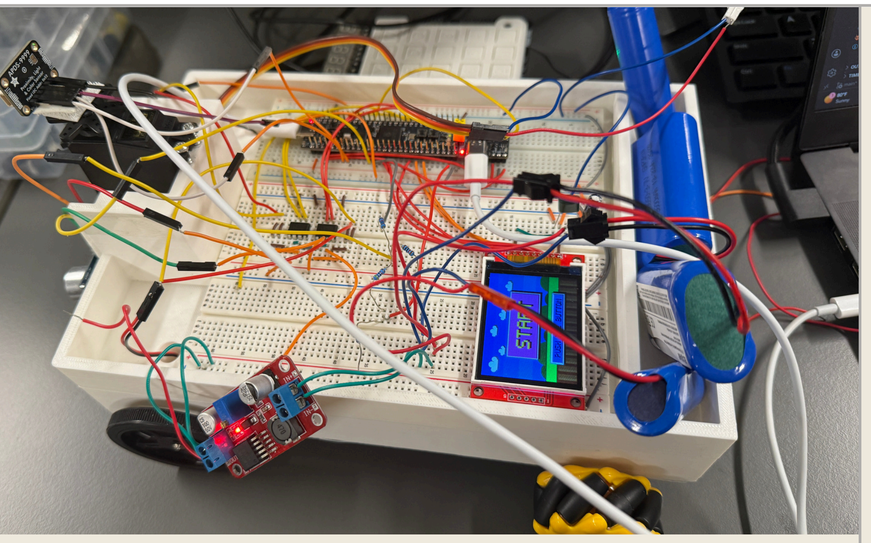
Color and Distance Sensor Robot (Group 26)

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Hardware Components:

- Adafruit APDS9999 Proximity, Lux, Light, and Color Sensor
- Ultrasonic Distance Sensor (HC-SR04 compatible)
- TowerPro SG92R Micro Servo
- L9110H H-Bridge Motor Driver
- Mecanum Wheels
- Lithium Ion Battery Pack
- TFT LCD display
- Push button for reset

ECE 362

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Project Overview:

The Color and Distance Sensor Robot is an autonomous robot that scans a 180-degree area to detect a target color, turns toward that target, moves in its direction, and stops once it reaches a set distance. The system combines multiple embedded interfaces and peripherals, including I2C, SPI, PWM, and ADC, to control sensors, motors, and display output.

Github Repository

Interfaces Utilization:

- I2C: sensor communication
- SPI: LCD/TFT display communication
- PWM: motor speed control and steering
- ADC: light intensity and battery voltage readings
- GPIO: reset button and control signals

How The Robot Works:

- The robot begins by searching for a human input of which color to search for.
- After the color is selected the servo with reset to the forward-facing state (0 degrees) and the color sensor will calibrate.
- The servo will begin scanning right to left searching for the color and stop the servo once the color is found.
- The robot then turns based on the degrees of the servo. Right if the degrees is positive and left is the degrees is negative.
- The robot will begin to go forward and approach the designated object and stop when the distance sensor detects that the object is less than 2.1 inches away.
- The motor will stop then the distance, color, and battery voltage will print on the TFT LCD.

Physical Implementations:

- Plain Wreckage Reconnaissance
- Space Rover
- Disaster Recovery Robot
- Sorting System
- Collision Prevention System